

$$④ L = \{a^i b^j c^k \mid k \neq i+j\}$$

$k < i+j \quad k > i+j$

$$S \rightarrow P \mid Q$$

$$P \rightarrow A \mid B \mid AB \mid W$$

$$W \rightarrow AWc \mid X \mid a$$

$$X \rightarrow BXc \mid b$$

$$Q \rightarrow C \mid R$$

$$R \rightarrow aRC \mid Y \mid c$$

$$Y \rightarrow bYC \mid c$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

$$C \rightarrow Cc \mid c$$

Paso 2: Eliminar proyecciones con λ

FÁCIL, NO HAY

$$S \rightarrow Aa \mid Bb \mid Cc \mid AWc \mid BXc \mid aRC \mid bYC \mid a \mid b \mid c$$

$$P \rightarrow Aa \mid Bb \mid AWc \mid BXc \mid b \mid a$$

$$W \rightarrow AWc \mid BXc \mid b \mid a$$

$$X \rightarrow BXc \mid b$$

$$Q \rightarrow Cc \mid aRC \mid bYC \mid c$$

$$R \rightarrow aRC \mid bYC \mid c$$

$$Y \rightarrow bYC \mid c$$

$$A \rightarrow Aa \mid a$$

$$B \rightarrow Bb \mid b$$

$$C \rightarrow Cc \mid c$$

Paso 1: Eliminar símbolos inútiles

Todos son generadores

$$\begin{cases} G_0 = \{W, X, R, Y, A, B, C\} \\ G_1 = \{P, Q, W, X, R, Y, A, B, C\} \\ G_2 = S + G_1 \\ G_3 = G_2 \end{cases}$$

Todos son alcanzables

$$\begin{cases} A_0 = \{S\} \\ A_1 = A_0 + \{P, Q\} \\ A_2 = A_1 + \{W, R, A, B, C\} \\ A_3 = A_2 + \{X, Y\} \\ A_4 = A_3 \end{cases}$$

Paso 3: Eliminar proyecciones unitarias

(S, S)	$(S, P)(S, Q)$	$(S, C)(S, R)$	$(S, A)(S, B)(S, W)$	(S, W)	(S, Y)
(P, P)	$(P, A)(P, B)(P, W)$	(P, W)			
(W, W)	(W, X)				
(X, X)					
(Q, Q)	$(Q, C)(Q, R)$	(Q, Y)			
(R, R)	(R, Y)				
(Y, Y)					
	L_1	L_2	L_3		